

MLE5217 – Take Home Assignments

Important!

There is no one correct answer.



You will be evaluated based on **how you critically analyse the materials science problem, make decisions on how to proceed with the task and explain your decisions and thinking!**

Important! : *#Comments*

- Write clear comments in the code so that another person can follow the logic.
- In instances where you have made decisions - justify them.
- Whenever, you have attempted more than one possibility, leave the original attempts (if possible) - don't delete or overwrite (you may comment them if needed).
- Acknowledge (if any) references or AI tools used at the bottom of the notebook.

Evaluation (Note that the allocated marks are tentative)

The primary emphasis will be on the depth and thoroughness of your approach to the problem. Key areas of focus will include:

- * Data Exploration: Demonstrating a thorough investigation of the data, exploring different analytical possibilities, and thoughtfully selecting the best course of action. (~7 marks)
- * Implementation: Translating your chosen approach into clean and efficient code. (~5 marks)
- * Machine Learning Process: Executing the machine learning process correctly and methodically, ensuring proper data handling, model selection, and evaluation. (~5 marks)
- * Clarity of Explanation: Providing clear explanations of each step, with logical reasoning for the decisions made. (~10 marks)
- * Critical Analysis: Identifying any limitations of the approach, suggesting potential improvements, and making relevant statistical inferences based on the results. (~8 marks)

Submission

Ensure that

- each of the cells of code in the final Jupyter notebooks have been Run for output (Except for the hyper-parameter optimization if any).
- Name the file by your name as "YourName.ipynb".
- It is your responsibility to Ensure that the correct files are being submitted, and the file extensions are in the correct format (.ipynb).
- Submission will be via Canvas, and late submissions will be penalized.

Assignment 1 (35 marks)

Dataset: Assign_regression_data.csv

Contains all non-metallic compounds with non-zero bandgap energies [Eg (eV)].

Also, contains electronegativity data;

en_min : the minimum electronegativity value of the elements in the compound

en_max : the maximum electronegativity value of the elements in the compound

en_weighted_mean : the weighted mean of the electronegativity values

en_spread : en_max – en_min

en_weighted_std : how spread out the electronegativities of the elements are around the mean.

Objective

 Fit a Regression Equation to the non-metallic compounds to predict the bandgap energies based on their chemical composition.

- Use a suitable composition-based feature vector to vectorize the chemical compounds. You may try multiple feature vectors and/or selected features/descriptors and analyse the outcomes.
 - In using a featurization scheme, note that you also have the choice to pick and choose your features.
 - Additional step: If you think is necessary, you may choose to include the electronegativity data that is given as descriptors (features), as electronegativity values are known to have an indirect relationship with bandgap.
- You may experiment with different models for regression analysis if required.
- Comment on the overall performance of the model and suggest any short-comings or potential improvements.

